Aadil Ali Mahar

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github.com/Adilmhr1262 · aadil-mahar-portfolio-v2.vercel.app

EDUCATION

B.S. Computer Science — The Islamia University of Bahawalpur, Pakistan 2022 – 2026

CGPA 3.02 · Graduated February 2026

Final-year project: *Health and Care* mobile health application (Grade A) — complete SRS and SDD design dossier: ambulance dispatch, hospital directory, and health records.

RESEARCH PROJECTS

Flood-Risk Prediction and Early-Warning Decision Support System 2026

- Machine-learning system for district-level flood risk prediction in Sindh, Pakistan.
- Trained and compared three models; selected an MLP neural network — F1 0.875, ROC-AUC 0.963 on held-out data covering 12 districts and 52 flood events.
- FastAPI + PostgreSQL/PostGIS backend, GIS risk layers (Leaflet), automated alert pipeline, Next.js public portal, and a Flutter mobile prototype.

Karachi Urban Heat & Green Cover Observatory 2026

- Reproducible satellite-data study relating vegetation (NDVI) to land surface temperature across Karachi, 2015–2025, from 176 USGS Landsat scenes.
- Pooled analysis (n = 12,563 grid cells): a 0.1 rise in NDVI associated with about 1.1 °C lower surface temperature (95% CI −12.54 to −10.03 °C per NDVI unit; R^2 0.40).
- Pre-registered research protocol, versioned data pipeline, and a full analytical report with diagnostics and stated limitations.

PALMS — Personalized Adaptive Learning Management System 2026

- Mastery-based learning platform with rule-based adaptive recommendations (five explicit rules, 70% mastery threshold) and auto-graded assessments.
- FastAPI + PostgreSQL backend, Next.js frontend, JWT cookie auth, Docker Compose deployment; 84-test suite passing.

Personal Nutrition & Calorie Tracking System 2025

- Web application for meal logging, food-database search via the Nutritionix API, personalized goals, and Chart.js progress dashboards, with a photo-based food identification flow.
- React frontend, Python Flask backend, SQLite.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Kotlin (design)
Frameworks	FastAPI, React, Next.js, Flask, Flutter (prototype)
Data & ML	scikit-learn, XGBoost, PostgreSQL/PostGIS, QGIS, Leaflet, Landsat data processing
Tools	Docker, Git, Alembic, Vercel

PROFESSIONAL EXPERIENCE

HR Specialist — Smart Marketing Methods, Germany (remote) · Feb 2025 – Sep 2026

Customer Service Representative — LIBT Admission Centre, UK (remote) · Apr 2025 – Sep 2025

Data Scientist — Data Science China, China (remote) · Oct 2024 – Mar 2025

Travel Agent — Asaptickets, USA (remote) · Feb 2024 – Jul 2024